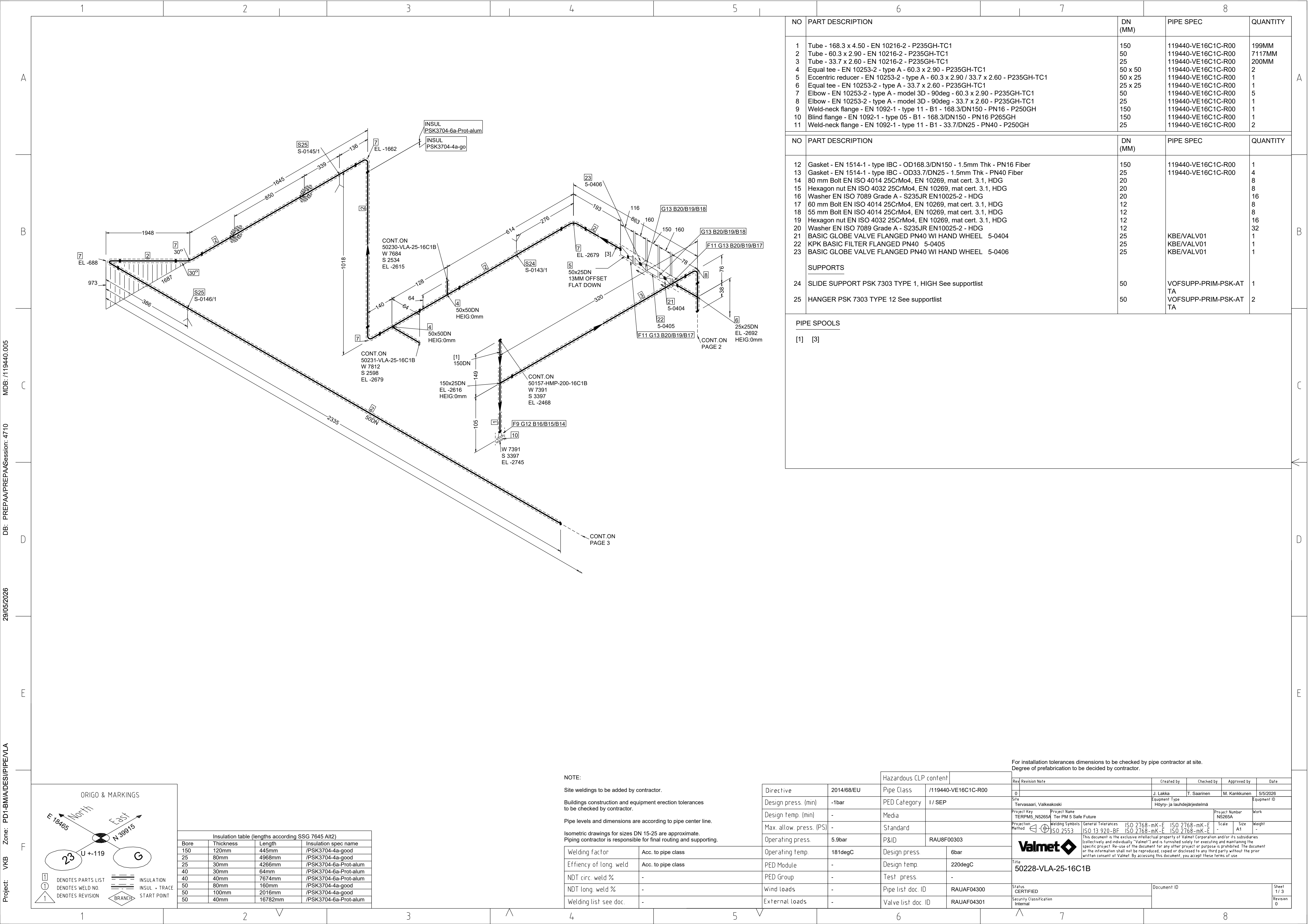
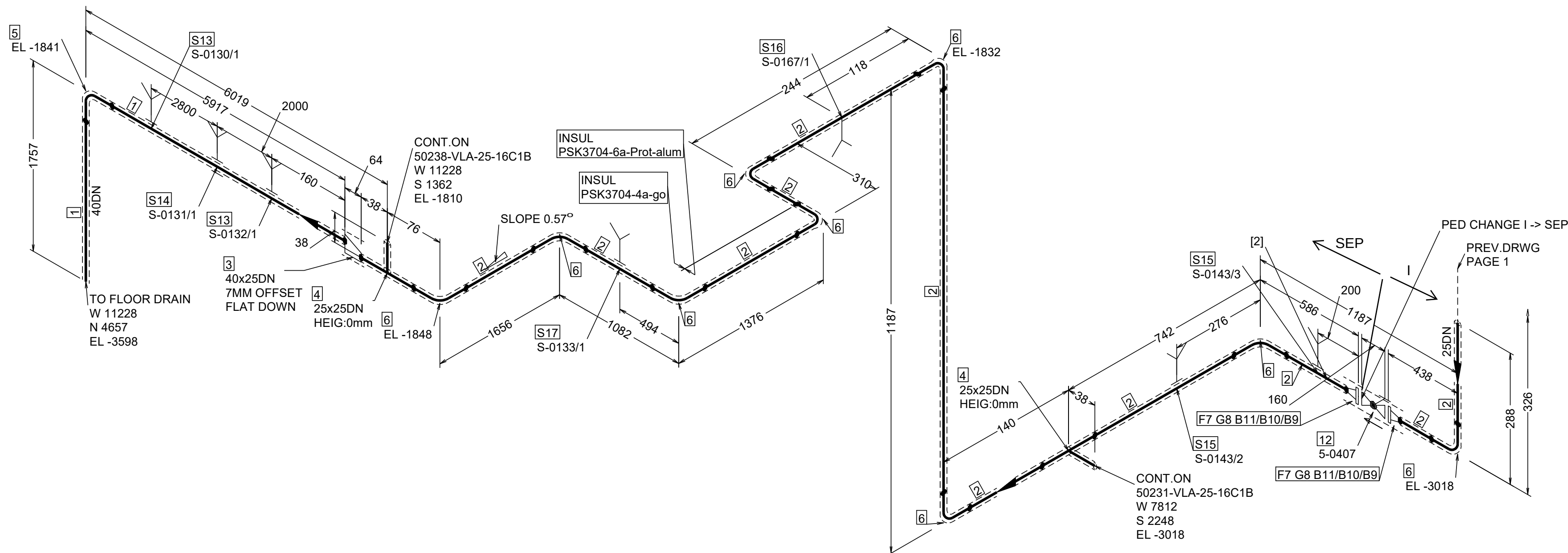
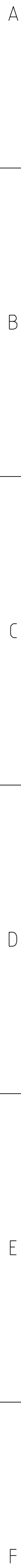




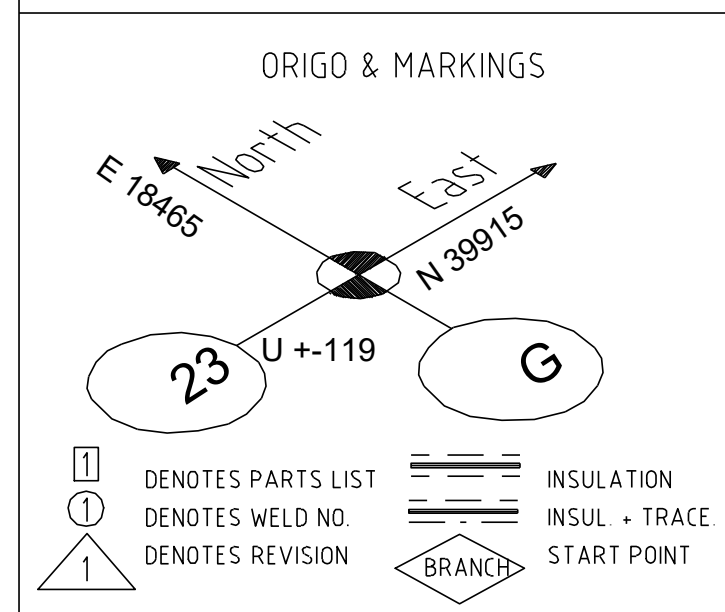




| NO | PART DESCRIPTION                                                                 | DN (MM) | PIPE SPEC              | QUANTITY |
|----|----------------------------------------------------------------------------------|---------|------------------------|----------|
| 1  | Tube - 48.3 x 2.60 - EN 10216-2 - P235GH-TC1                                     | 40      | 119440-VE16C1C-R00     | 7560MM   |
| 2  | Tube - 33.7 x 2.60 - EN 10216-2 - P235GH-TC1                                     | 25      | 119440-VE16C1C-R00     | 7247MM   |
| 3  | Eccentric reducer - EN 10253-2 - type A - 48.3 x 2.60 / 33.7 x 2.60 - P235GH-TC1 | 40 x 25 | 119440-VE16C1C-R00     | 1        |
| 4  | Equal tee - EN 10253-2 - type A - 33.7 x 2.60 - P235GH-TC1                       | 25 x 25 | 119440-VE16C1C-R00     | 2        |
| 5  | Elbow - EN 10253-2 - type A - model 3D - 90deg - 48.3 x 2.60 - P235GH-TC1        | 40      | 119440-VE16C1C-R00     | 1        |
| 6  | Elbow - EN 10253-2 - type A - model 3D - 90deg - 33.7 x 2.60 - P235GH-TC1        | 25      | 119440-VE16C1C-R00     | 9        |
| 7  | Weld-neck flange - EN 1092-1 - type 11 - B1 - 33.7/DN25 - PN40 - P250GH          | 25      | 119440-VE16C1C-R00     | 2        |
| NO | PART DESCRIPTION                                                                 | DN (MM) | PIPE SPEC              | QUANTITY |
| 8  | Gasket - EN 1514-1 - type IBC - OD33.7/DN25 - 1.5mm Thk - PN40 Fiber             | 25      | 119440-VE16C1C-R00     | 2        |
| 9  | 60 mm Bolt EN ISO 4014 25CrMo4, EN 10269, mat cert. 3.1, HDG                     | 12      |                        | 8        |
| 10 | Hexagon nut EN ISO 4032 25CrMo4, EN 10269, mat cert. 3.1, HDG                    | 12      |                        | 8        |
| 11 | Washer EN ISO 7089 Grade A - S235JR EN10025-2 - HDG                              | 12      |                        | 16       |
| 12 | BASIC GLOBE VALVE FLANGED PN40 WI HAND WHEEL 5-0407                              | 25      | KBE/VALV01             | 1        |
|    | SUPPORTS                                                                         |         |                        |          |
| 13 | SLIDE SUPPORT PSK 7303 TYPE 1, LOW See supportlist                               | 40      | VOFSUPP-PRIM-PSK-AT TA | 2        |
| 14 | GUIDED SLIDE PSK 7303 TYPE 2, LOW See supportlist                                | 40      | VOFSUPP-PRIM-PSK-AT TA | 1        |
| 15 | SLIDE SUPPORT PSK 7303 TYPE 1, HIGH See supportlist                              | 25      | VOFSUPP-PRIM-PSK-AT TA | 2        |
| 16 | HANGER PSK 7303 TYPE 12 See supportlist                                          | 25      | VOFSUPP-PRIM-PSK-AT TA | 1        |
| 17 | SLIDE SUPPORT PSK 7303 TYPE 1, LOW See supportlist                               | 25      | VOFSUPP-PRIM-PSK-AT TA | 1        |

## PIPE SPOOLS

[2]



| Insulation table (lengths according SSG 7645 AlP2) |           |         |                       |
|----------------------------------------------------|-----------|---------|-----------------------|
| Bore                                               | Thickness | Length  | Insulation spec name  |
| 150                                                | 120mm     | 445mm   | /PSK3704-4a-good      |
| 25                                                 | 80mm      | 496mm   | /PSK3704-4a-good      |
| 25                                                 | 30mm      | 4266mm  | /PSK3704-6a-Prot-alum |
| 40                                                 | 30mm      | 64mm    | /PSK3704-6a-Prot-alum |
| 40                                                 | 40mm      | 7674mm  | /PSK3704-6a-Prot-alum |
| 50                                                 | 80mm      | 160mm   | /PSK3704-4a-good      |
| 50                                                 | 100mm     | 2016mm  | /PSK3704-4a-good      |
| 50                                                 | 40mm      | 16782mm | /PSK3704-6a-Prot-alum |

NOTE:

Site weldings to be added by contractor.

Buildings construction and equipment erection tolerances to be checked by contractor.

Pipe levels and dimensions are according to pipe center line.

Isometric drawings for sizes DN 15-25 are approximate.  
Piping contractor is responsible for final routing and supporting.

|                         |                    |
|-------------------------|--------------------|
| Welding factor          | Acc. to pipe class |
| Efficiency of long weld | Acc. to pipe class |
| NDT circ. weld %        | -                  |
| NDT long weld %         | -                  |
| Welding list see doc.   | -                  |

For installation tolerances dimensions to be checked by pipe contractor at site.  
Degree of prefabrication to be decided by contractor.

|                         |            |                       |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  |  |                                                                               |  |             |  |                          |  |              |  |             |  |
|-------------------------|------------|-----------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|-------------------------------------------------------------------------------|--|-------------|--|--------------------------|--|--------------|--|-------------|--|
|                         |            | Hazardous CLP content |                     | Figure 1: Product information to be included by customer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |                                                                               |  |             |  |                          |  |              |  |             |  |
| Directive               | 2014/68/EU | Pipe Class            | /I19440-VE16C1C-R00 | Rev/ Revision Note                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |  | Created by                                                                    |  | Checked by  |  | Approved by              |  | Date         |  |             |  |
| Design press. (min)     | -1bar      | PED Category          | I / SEP             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |  | J. Lakka                                                                      |  | T. Saarinen |  | M. Kankkunen             |  | 5/5/2026     |  |             |  |
| Design temp. (min)      | -          | Media                 |                     | Site<br>Tervasaari, Vaikoeoski                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |  | Equipment Type<br>Höyry- ja lauhdejärjestelmä                                 |  |             |  |                          |  | Equipment ID |  |             |  |
| Max. allow. press. (PS) | -          | Standard              |                     | Project Key<br>TERPMS_NS265A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |  | Project Name<br>Ter PM 5 Safe Future                                          |  |             |  | Project Number<br>NS265A |  | Work         |  |             |  |
| Operating press.        | 5.9bar     | P&ID                  | RAUF00303           | Welding Symbols<br>ISO 2553                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  | General Tolerances<br>ISO 13 920-BF ISO 2768-mK-E ISO 2768-mK-E ISO 2768-mK-E |  |             |  | Scale<br>-               |  | Size<br>A1   |  | Weight<br>- |  |
| Operating temp.         | 181degC    | Design press          | 6bar                | <div></div> <p>This document is the exclusive intellectual property of Valmet Corporation and/or its subsidiaries (collectively and individually "Valmet") and is furnished solely for executing and maintaining the specific project. Re-use of the document for any other project or purpose is prohibited. The document or the information shall not be reproduced, copied or disclosed to any third party without the prior written consent of Valmet. By accessing this document, you accept these terms of use.</p> |  |  |  |                                                                               |  |             |  |                          |  |              |  |             |  |
| PED Module              | -          | Design temp.          | 220degC             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  |  |                                                                               |  |             |  |                          |  |              |  |             |  |
| PED Group               | -          | Test press.           | -                   | Title<br>50228-VLA-25-16C1B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                                                                               |  |             |  |                          |  |              |  |             |  |
| Wind loads              | -          | Pipe list doc. ID     | RAUAF04300          | Status<br>CERTIFIED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  |  | Document ID                                                                   |  |             |  | Sheet<br>2 / 3           |  |              |  |             |  |
| External loads          | -          | Valve list doc. ID    | RAUAF04301          | Security Classification<br>Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  |  |                                                                               |  |             |  | Revision<br>0            |  |              |  |             |  |



BD

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